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18/557,529	10/26/2023	Francesco GATTO	SYNP101US-TJC	9196

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EXAMINER	
WECKER, JENNIFER	

ART UNIT	PAPER NUMBER
1797	

NOTIFICATION DATE	DELIVERY MODE
06/18/2026	ELECTRONIC

Please find below and/or attached an Office communication concerning this application or proceeding.

The time period for reply, if any, is set in the attached communication.

Notice of the Office communication was sent electronically on above-indicated "Notification Date" to the following e-mail address(es):

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Office Action Summary	Application No. 18/557,529	Applicant(s) GATTO et al.	
	Examiner JENNIFER WECKER	Art Unit 1797	AIA (First Inventor to File) Status Yes

-- The MAILING DATE of this communication appears on the cover sheet with the correspondence address --

Period for Reply

A SHORTENED STATUTORY PERIOD FOR REPLY IS SET TO EXPIRE 3 MONTHS FROM THE MAILING DATE OF THIS COMMUNICATION.

- Extensions of time may be available under the provisions of 37 CFR 1.136(a). In no event, however, may a reply be timely filed after SIX (6) MONTHS from the mailing date of this communication.
- If NO period for reply is specified above, the maximum statutory period will apply and will expire SIX (6) MONTHS from the mailing date of this communication.
- Failure to reply within the set or extended period for reply will, by statute, cause the application to become ABANDONED (35 U.S.C. § 133). Any reply received by the Office later than three months after the mailing date of this communication, even if timely filed, may reduce any earned patent term adjustment. See 37 CFR 1.704(b).

Status

1)☑ Responsive to communication(s) filed on 06/02/2026.

☐ A declaration(s)/affidavit(s) under **37 CFR 1.130(b)** was/were filed on _____.

2a)☐ This action is **FINAL.** 2b) ☑ This action is non-final.

3)☐ An election was made by the applicant in response to a restriction requirement set forth during the interview on _____; the restriction requirement and election have been incorporated into this action.

4)☐ Since this application is in condition for allowance except for formal matters, prosecution as to the merits is closed in accordance with the practice under *Ex parte Quayle*, 1935 C.D. 11, 453 O.G. 213.

Disposition of Claims*

5) ☑ Claim(s) 1-33 and 35-36 is/are pending in the application.

5a) Of the above claim(s) 33 and 35-36 is/are withdrawn from consideration.

6) ☐ Claim(s) _____ is/are allowed.

7) ☑ Claim(s) 1-31 is/are rejected.

8) ☐ Claim(s) _____ is/are objected to.

9) ☐ Claim(s) _____ are subject to restriction and/or election requirement

* If any claims have been determined allowable, you may be eligible to benefit from the **Patent Prosecution Highway** program at a participating intellectual property office for the corresponding application. For more information, please see http://www.uspto.gov/patents/init_events/pph/index.jsp or send an inquiry to **PPHfeedback@uspto.gov.**

Application Papers

10)☐ The specification is objected to by the Examiner.

11)☑ The drawing(s) filed on 10/26/2023 is/are: a)☑ accepted or b)☐ objected to by the Examiner.

Applicant may not request that any objection to the drawing(s) be held in abeyance. See 37 CFR 1.85(a).

Replacement drawing sheet(s) including the correction is required if the drawing(s) is objected to. See 37 CFR 1.121(d).

Priority under 35 U.S.C. § 119

12)☑ Acknowledgment is made of a claim for foreign priority under 35 U.S.C. § 119(a)-(d) or (f).

Certified copies:

a)☑ All b)☐ Some** c)☐ None of the:

1.☐ Certified copies of the priority documents have been received.

2.☐ Certified copies of the priority documents have been received in Application No. _____.

3.☑ Copies of the certified copies of the priority documents have been received in this National Stage application from the International Bureau (PCT Rule 17.2(a)).

** See the attached detailed Office action for a list of the certified copies not received.

Attachment(s)

1) ☑ Notice of References Cited (PTO-892)

2) ☑ Information Disclosure Statement(s) (PTO/SB/08a and/or PTO/SB/08b) Paper No(s)/Mail Date _____.

3) ☐ Interview Summary (PTO-413) Paper No(s)/Mail Date _____.

4) ☐ Other: _____.

DETAILED ACTION

Notice of Pre-AIA or AIA Status

The present application, filed on or after March 16, 2013, is being examined under the first inventor to file provisions of the AIA.

Election/Restrictions

Claims 32-33 and 35-36 are withdrawn from further consideration pursuant to 37 CFR 1.142(b) as being drawn to a nonelected invention, there being no allowable generic or linking claim. Election was made **without** traverse in the reply filed on 06/02/2026.

Therefore, Applicant's election without traverse of Group I, claims 1-31, in the reply filed on 6/2/2026 is acknowledged.

Claim Rejections - 35 USC § 102

The following is a quotation of the appropriate paragraphs of 35 U.S.C. 102 that form the basis for the rejections under this section made in this Office action:

> A person shall be entitled to a patent unless –
>
> (a)(1) the claimed invention was patented, described in a printed publication, or in public use, on sale, or otherwise available to the public before the effective filing date of the claimed invention.

Claim(s) 1-26 and 29-31 are rejected under 35 U.S.C. 102(a)(1) as being anticipated by Gatto et al (US PGPub 2019/0391151, cited in the IDS).

Regarding Claim 1, Gatto et al teaches a method of screening for cancer in a subject (see abstract), said method comprising determining the level and/or chemical

composition of the protein-free fraction of one or both of the glycosaminoglycans (GAGs) chondroitin sulfate (CS) and heparan sulfate (HS) in a body fluid sample (see [0042]-[0044] and [0583], which discloses that the CS and/or HS are separated (i.e. free) from a protein) , wherein said sample has been obtained from said subject (see [0006] and [0042]).

Regarding Claim 2, Gatto et al teaches that an altered level and/or chemical composition of chondroitin sulfate (CS) and/or heparan sulfate (HS) in said protein-free fraction in comparison to a control level and/or chemical composition is indicative of cancer in said subject (see [0040], [0093] and [0100]).

Regarding Claim 3, Gatto et al teaches that said determination of the chemical composition comprises determining the level in said protein-free fraction of one or more GAG properties selected from the group consisting of: one or more (i.e. all) of the specific sulfated or unsulfated forms of CS or HS disaccharides, charge HS, charge CS, the total concentration of CS or the total concentration of HS (see [0057]).

Regarding Claim 4, Gatto et al teaches that said determination of the chemical composition comprises determining the level in said protein-free fraction of one or more (or all) of the GAG properties selected from the group consisting of: the specific sulfated or unsulfated forms of CS or HS disaccharides selected from the group consisting of: Os CS, 2s CS, 6s CS, 4s CS, 2s6s CS, 2s4s CS, 4s6s CS,Tris CS, Os HS, 2s HS, 6s HS, 2s6s HS, Ns HS, Ns2s HS, Ns6s HS, Tris HS, the ratio of 4s CS to 6s CS, the ratio of 6s CS to Os CS and the ratio of 4s CS to Os CS; charge HS; charge CS; the total concentration of CS (CS tot); and the total concentration of HS (HS tot) (see [0081]-[0087]).

Regarding Claim 5, Gatto et al teaches that said body fluid sample is blood (e.g. plasma) and/or urine (see [0047], [0192] and [0583]).

Regarding Claim 6, Gatto et al teaches that said determination of the chemical composition comprises determining the level in said protein-free fraction of (i) one or more (or all) of the GAG properties selected from the group consisting of: the specific sulfated or unsulfated forms of CS or HS disaccharides selected from the group consisting of: the specific sulfated or unsulfated forms of CS or HS disaccharides selected from the group consisting of: absolute concentration of 4s CS, absolute concentration of Os CS, relative concentration of Os CS, absolute concentration of Os HS, absolute concentration of Ns HS, absolute concentration of 6s CS, absolute concentration of 2s6s CS, relative concentration of 4s CS, relative concentration of 2s6s CS, relative concentration of 6s CS, relative concentration of Os HS, relative concentration of Ns HS; the ratio of 6s CS to Os CS; the ratio of 4s CS to Os CS; CS tot; HS tot; charge CS; or (ii) one or more (or all) of the GAG properties selected from the group consisting of: the specific sulfated or unsulfated forms of CS or HS disaccharides selected from the group consisting of: absolute concentration of 4s CS, absolute concentration of Os CS, relative concentration of Os CS, absolute concentration of Os HS, absolute concentration of Ns HS, absolute concentration of 6s CS, absolute concentration of 2s6s CS, relative concentration of 4s CS, relative concentration of 2s6s CS, relative concentration of 6s CS; CS tot; HS tot; charge CS; or (iii) one or more (or all) of the GAG properties selected from the group consisting of: the specific sulfated or unsulfated forms of CS or HS disaccharides selected from the group consisting of: absolute concentration of 4s CS, absolute concentration of Os CS,

relative concentration of Os CS, absolute concentration of Os HS, absolute concentration of Ns HS, absolute concentration of 2s6s CS, relative concentration of 4s CS, relative concentration of 2s6s CS, relative concentration of 6s CS, relative concentration of Os HS, relative concentration of Ns HS; the ratio of 6s CS to Os CS; the ratio of 4s CS to Os CS; CS tot; HS tot; charge CS (see [0074]-[0075] and [0082]).

Regarding Claim 7, Gatto et al teaches that said determination of the chemical composition comprises determining the level in said protein-free fraction of one or more (or all) of the GAG properties selected from the group consisting of: the specific sulfated or unsulfated forms of CS or HS disaccharides selected from the group consisting of: absolute concentration of 4s CS, absolute concentration of Os CS, relative concentration of Os CS, absolute concentration of Os HS, absolute concentration of Ns HS, absolute concentration of 2s6s CS, relative concentration of 4s CS, relative concentration of 2s6s CS, relative concentration of 6s CS, relative concentration of Os HS, relative concentration of Ns HS, absolute concentration of 6s CS; the ratio of 6s CS to Os CS; the ratio of 4s CS to Os CS; CS tot; HS tot; charge CS (see [0081]-[0095]).

Regarding Claim 8, Gatto et al teaches that said body fluid sample is blood (e.g. plasma) and said determination of the chemical composition comprises determining the level in said protein-free fraction of (i) one or more (or all) of the GAG properties selected from the group consisting of: the specific sulfated or unsulfated forms of CS or HS disaccharides selected from the group consisting of: absolute concentration of Os CS, relative concentration of Os CS, absolute concentration of 4s CS, relative concentration of 4s CS; the ratio of 4s CS to Os CS; CS Tot; charge CS; or (ii) one or more (or all) of the GAG properties selected from the group consisting of: the specific

sulfated or unsulfated forms of CS or HS disaccharides selected from the group consisting of: absolute concentration of Os CS, relative concentration of Os CS, absolute concentration of 4s CS, relative concentration of 4s CS; CS Tot; charge CS; or (iii) one or more (or all) of the GAG properties selected from the group consisting of: the specific sulfated or unsulfated forms of CS or HS disaccharides selected from the group consisting of: absolute concentration of Os CS, relative concentration of 4s CS; the ratio of 4s CS to Os CS (see [0081]-[0095]).

Regarding Claim 9, Gatto et al teaches that said body fluid sample is urine and said determination of the chemical composition comprises determining the level in said protein-free fraction of (i) one or more (or all) of the GAG properties selected from the group consisting of: the specific sulfated or unsulfated forms of CS or HS disaccharides selected from the group consisting of: absolute concentration of 4s CS, absolute concentration of Os CS, relative concentration of Os CS, absolute concentration of Os HS, absolute concentration of Ns HS, absolute concentration of 6s CS, absolute concentration of 2s6s CS, relative concentration of 4s CS, relative concentration of 2s6s CS, relative concentration of 6s CS, relative concentration of Os HS, relative concentration of Ns HS; the ratio of 6s CS to Os CS; CS tot; HS tot; charge CS; or (ii) one or more (or all) of the GAG properties selected from the group consisting of: the specific sulfated or unsulfated forms of CS or HS disaccharides selected from the group consisting of: absolute concentration of 4s CS, absolute concentration of Os CS, relative concentration of Os CS, absolute concentration of Os HS, absolute concentration of Ns HS, absolute concentration of 6s CS, absolute concentration of 2s6s CS, relative concentration of 2s6s CS, relative concentration of 6s CS; CS tot; HS

tot; charge CS; or (iii) one or more (or all) of the GAG properties selected from the group consisting of: the specific sulfated or unsulfated forms of CS or HS disaccharides selected from the group consisting of: absolute concentration of 4s CS, relative concentration of Os CS, absolute concentration of Os HS, absolute concentration of Ns HS, absolute concentration of 2s6s CS, relative concentration of 4s CS, relative concentration of 2s6s CS, relative concentration of 6s CS, relative concentration of Os HS, relative concentration of Ns HS; the ratio of 6s CS to Os CS; CS tot; HS tot (see [0074], [0082], [0088]-[0089] and [0133]).

Regarding Claim 10, Gatto et al teaches that said body fluid sample is urine and said determination of the chemical composition comprises determining the level in said protein- free fraction of one or more (or all) of the GAG properties selected from the group consisting of: the specific sulfated or unsulfated forms of CS or HS disaccharides selected from the group consisting of: absolute concentration of Os HS, relative concentration of 6s CS, relative concentration of Ns HS, relative concentration of Os HS, relative concentration of 2s6s CS, relative concentration of 4s CS, relative concentration of Os CS, absolute concentration of 4s CS, absolute concentration of 6s CS, absolute concentration of Os CS; the ratio of 6s CS to Os CS; CS tot; HS tot (see [0074], [0082], [0088]-[0089] and [0133]).

Regarding Claim 11, Gatto et al teaches the chemical composition of said one or both of said GAGs in said protein-free fraction of a blood (e.g. plasma) sample is determined (see [0074] and [0084]) and the chemical composition of said one or both of said GAGs in said protein-free fraction of a urine sample is performed, preferably wherein determination of the chemical composition of said protein-free fraction of said

urine sample comprises determining the level in said protein-free fraction of one or more (or all) of the GAG properties selected from the group consisting of: relative concentration of Os CS, absolute concentration of Os HS, absolute concentration of Ns HS, relative concentration of 2s6s CS, relative concentration of Os HS, relative concentration of Ns HS; HS tot; and charge CS (see [0081]-[0095]); and wherein determination of the chemical composition of said protein-free fraction of said blood (e.g. plasma) sample comprises determining the level in said protein-free fraction of one or more (or all) of the GAG properties selected from the group consisting of: absolute concentration of Os CS; relative concentration of 4s CS; and CS tot (see [0082] and [0124]).

Regarding Claim 12, Gatto et al teaches that the chemical composition of said one or both of said GAGs in said protein-free fraction of a blood (e.g. plasma) sample is determined and the chemical composition of said one or both of said GAGs in said protein-free fraction of a urine sample is performed (see [0074] and [0084]), wherein determination of the chemical composition of said protein-free fraction of said urine sample comprises determining the level in said protein-free fraction of one or more (or all) of the GAG properties selected from the group consisting of: absolute concentration of Os HS, relative concentration of Ns HS, relative concentration of Os CS, relative concentration of Os HS, absolute concentration of 4s CS, absolute concentration of Os CS, relative concentration of 4s CS, relative concentration of 6s CS, absolute concentration of 6s CS; the ratio of 6s CS to Os CS; CS tot; and charge CS (see [0081]-[0095]); and wherein determination of the chemical composition of said protein-free fraction of said blood (e.g. plasma) sample comprises determining the level in said

protein-free fraction of one or both of the GAG properties selected from the group consisting of: the ratio of 4s CS to Os CS; and CS tot (see [0082]-[0084], [0098], [0124] and [0162]).

Regarding Claim 13, Gatto et al teaches that the chemical composition of said one or both of said GAGs in said protein-free fraction of a blood (e.g. plasma) sample is determined and the chemical composition of said one or both of said GAGs in said protein-free fraction of a urine sample is performed (see [0074] and [0084]), wherein determination of the chemical composition of said protein-free fraction of said urine sample comprises determining the level in said protein-free fraction of one or more (or all) of the GAG properties selected from the group consisting of: absolute concentration of Os HS, absolute concentration of Ns HS, absolute concentration of 4s CS, absolute concentration of 6s CS (see [0081]-[0095]); and wherein determination of the chemical composition of said protein-free fraction of said blood (e.g. plasma) sample comprises determining the level in said protein-free fraction of absolute concentration of Os CS (see [0075]).

Regarding Claim 14, Gatto et al teaches that determination of the level and/or chemical composition of said protein-free fraction of said body fluid sample comprises determining the level in said protein-free fraction of one or both of the GAG properties selected from the group consisting of: the absolute concentration of Os CS and CS Tot (see [0075], [0082]-[0084], [0098], [0124] and [0162]).

Regarding Claim 15, Gatto et al teaches determining the level in said protein-free fraction of the absolute concentration of Os CS and wherein an increase in the absolute

concentration of Os CS, in comparison to a control level, is indicative of cancer (see [0075] and [0093]).

Regarding Claim 16, Gatto et al teaches that the chemical composition may be expressed in terms of score, said score being based on the measured level of one or more (preferably more than one) or all of said GAG properties (see [0195] and [0198]).

Regarding Claim 17, Gatto et al teaches determining the level of more than one of said GAG properties, preferably said method comprises determining the level of two or more, three or more, four or more, or all, of said GAG properties (see [0042] and [0143]).

Regarding Claims 18 and 20, Gatto et al teaches that said sample has been obtained from said subject and has been subjected to at least one processing step prior to determining said level and/or chemical composition, and wherein said at least one processing step does not comprise contacting said sample with a proteolytic agent (instead involving a step of fragmentation or cleavage or digestion, e.g. by chemical digestion or enzyme treatment, e.g. with chondroitinase ABC or chondroitinase B) (see [0077] and [0583]).

Regarding Claim 19, Gatto et al teaches that the levels of one or more of the specific sulfated or unsulfated forms of CS or HS disaccharides are determined, and wherein the GAGs are, or have been, subjected to a processing step to obtain the disaccharide units for analysis (see [0077] and [0584]).

Regarding Claims 21-22, Gatto et al teaches that said sample has been obtained from said subject and has been subjected to processing prior to determining said level and/or chemical composition, wherein said processing (a) comprises fragmenting said

one or both GAGs into disaccharide units; and (b) does not comprise prior to (a) at least one of:(i) contacting said sample with a proteolytic agent (see [0077], [0583]-[0584] and [0586]).

Regarding Claim 23-24, Gatto et al teaches that said fragmenting of (a) is performed by contacting said one or both GAGs with one or more GAG lyase enzymes, such as chondrotinase enzymes (see [0584] and [0722]).

Regarding Claim 25, Gatto et al teaches that said purifying of (b)(ii) is performed using an anion-exchange resin (see [0078] and [0583]).

Regarding Claim 26, Gatto et al teaches that said level or chemical composition of said GAG or GAG property is determined by HPLC and mass spectrometry (see [0045], [0076] and [0575]).

Regarding Claims 29-30, Gatto et al teaches that said subject is a subject at risk of developing cancer, or at risk of the occurrence of cancer, or is a subject having or suspected of having cancer (see abstract and [0549]), and wherein said cancer is (i) a stage I cancer, a stage II cancer, a stage III cancer, a stage IV cancer and a cancer of an unspecified stage; or (ii) selected from the group consisting of a genitourinary cancer (e.g. kidney cancer, prostate cancer or bladder cancer), a respiratory tract cancer (e.g. lung cancer), a brain tumor, a blood cancer (e.g. a lymphoma), colorectal cancer, uterine cancer, a gastrointestinal- neuroendocrine tumour, a breast cancer, ovarian cancer and head and neck cancer; or (iii) selected from the group consisting of: bladder cancer, breast invasive ductal carcinoma, cervix squamous cell carcinoma, chronic lymphoid leukaemia, colorectal cancer, endometrial cancer, diffuse glioma, a gastro-intestinal endocrine tumour, head and neck squamous cell carcinoma, diffuse large B-

cell lymphoma, non-small cell lung cancer, ovarian cancer, prostate cancer and renal cell cancer (see [0010], [0549] and [0634]).

Regarding Claim 31, Gatto et al teaches said method is used for diagnosing cancer, for the prognosis of cancer, for predicting the occurrence of cancer, for estimate the risk of the occurrence of cancer, for monitoring subjects at risk of the occurrence of cancer, for monitoring the progression of cancer in a subject, for determining the clinical severity of cancer, for predicting the response of a subject to therapy or surgery for cancer, for determining the efficacy of a therapeutic or surgical regime being used to treat cancer, for detecting the recurrence of cancer, for predicting the tissue of origin of a cancer, or for distinguishing small masses suspicious of cancer from other non-malignant diseases (see [0217], [0245] and [0549]).

Claim(s) 1-7, 9-10, 26 and 29-31 are rejected under 35 U.S.C. 102(a)(1) as being anticipated by Crawford et al (US PGPub 2012/0009616).

Regarding Claim 1, Crawford et al teaches a method of screening for cancer in a subject (see [0115] and [0274]), said method comprising determining the level and/or chemical composition of the protein-free fraction of one or both of the glycosaminoglycans (GAGs) chondroitin sulfate (CS) and heparan sulfate (HS) in a body fluid sample , wherein said sample has been obtained from said subject (see abstract, [0041]-[0043] and [0099]).

Regarding Claim 2, Crawford et al teaches that an altered level and/or chemical composition of chondroitin sulfate (CS) and/or heparan sulfate (HS) in said protein-free

fraction in comparison to a control level and/or chemical composition is indicative of cancer in said subject (see [0042], [0052], [0120] and [0235]).

Regarding Claim 3, Crawford et al teaches that said determination of the chemical composition comprises determining the level in said protein-free fraction of one or more GAG properties selected from the group consisting of: one or more (i.e. all) of the specific sulfated or unsulfated forms of CS or HS disaccharides, charge HS, charge CS, the total concentration of CS or the total concentration of HS (specifically the total concentration of HS or CS (see [0311]-[0312]).

Regarding Claim 4, Crawford et al teaches that said determination of the chemical composition comprises determining the level in said protein-free fraction of one or more of the GAG properties selected from the group consisting of: the specific sulfated or unsulfated forms of CS or HS disaccharides selected from the group consisting of: Os CS, 2s CS, 6s CS, 4s CS, 2s6s CS, 2s4s CS, 4s6s CS,Tris CS, Os HS, 2s HS, 6s HS, 2s6s HS, Ns HS, Ns2s HS, Ns6s HS, Tris HS, the ratio of 4s CS to 6s CS, the ratio of 6s CS to Os CS and the ratio of 4s CS to Os CS; charge HS; charge CS; the total concentration of CS (CS tot); and the total concentration of HS (HS tot) (specifically the total concentration of HS or CS (see [0311]-[0312]).

Regarding Claim 5, Crawford et al teaches that said body fluid sample is blood (e.g. plasma) and/or urine (see [0102], [0106] and [0246]).

Regarding Claim 6, Gatto et al teaches that said determination of the chemical composition comprises determining the level in said protein-free fraction of (i) one or more (or all) of the GAG properties selected from the group consisting of: the specific sulfated or unsulfated forms of CS or HS disaccharides selected from the group

consisting of: the specific sulfated or unsulfated forms of CS or HS disaccharides selected from the group consisting of: absolute concentration of 4s CS, absolute concentration of Os CS, relative concentration of Os CS, absolute concentration of Os HS, absolute concentration of Ns HS, absolute concentration of 6s CS, absolute concentration of 2s6s CS, relative concentration of 4s CS, relative concentration of 2s6s CS, relative concentration of 6s CS, relative concentration of Os HS, relative concentration of Ns HS; the ratio of 6s CS to Os CS; the ratio of 4s CS to Os CS; CS tot; HS tot; charge CS; or (ii) one or more (or all) of the GAG properties selected from the group consisting of: the specific sulfated or unsulfated forms of CS or HS disaccharides selected from the group consisting of: absolute concentration of 4s CS, absolute concentration of Os CS, relative concentration of Os CS, absolute concentration of Os HS, absolute concentration of Ns HS, absolute concentration of 6s CS, absolute concentration of 2s6s CS, relative concentration of 4s CS, relative concentration of 2s6s CS, relative concentration of 6s CS; CS tot; HS tot; charge CS; or (iii) one or more (or all) of the GAG properties selected from the group consisting of: the specific sulfated or unsulfated forms of CS or HS disaccharides selected from the group consisting of: absolute concentration of 4s CS, absolute concentration of Os CS, relative concentration of Os CS, absolute concentration of Os HS, absolute concentration of Ns HS, absolute concentration of 2s6s CS, relative concentration of 4s CS, relative concentration of 2s6s CS, relative concentration of 6s CS, relative concentration of Os HS, relative concentration of Ns HS; the ratio of 6s CS to Os CS; the ratio of 4s CS to Os CS; CS tot; HS tot; charge CS (specifically the total concentration of HS or CS (see [0311]-[0312]).

Regarding Claim 7, Gatto et al teaches that said determination of the chemical composition comprises determining the level in said protein-free fraction of one or more of the GAG properties selected from the group consisting of: the specific sulfated or unsulfated forms of CS or HS disaccharides selected from the group consisting of: absolute concentration of 4s CS, absolute concentration of Os CS, relative concentration of Os CS, absolute concentration of Os HS, absolute concentration of Ns HS, absolute concentration of 2s6s CS, relative concentration of 4s CS, relative concentration of 2s6s CS, relative concentration of 6s CS, relative concentration of Os HS, relative concentration of Ns HS, absolute concentration of 6s CS; the ratio of 6s CS to Os CS; the ratio of 4s CS to Os CS; CS tot; HS tot; charge CS (specifically the total concentration of HS or CS (see [0311]-[0312]).

Regarding Claim 9, Crawford et al teaches that said body fluid sample is urine (see [0102] and [0106]) and said determination of the chemical composition comprises determining the level in said protein-free fraction of (i) one or more of the GAG properties selected from the group consisting of: the specific sulfated or unsulfated forms of CS or HS disaccharides selected from the group consisting of: absolute concentration of 4s CS, absolute concentration of Os CS, relative concentration of Os CS, absolute concentration of Os HS, absolute concentration of Ns HS, absolute concentration of 6s CS, absolute concentration of 2s6s CS, relative concentration of 4s CS, relative concentration of 2s6s CS, relative concentration of 6s CS, relative concentration of Os HS, relative concentration of Ns HS; the ratio of 6s CS to Os CS; CS tot; HS tot; charge CS; or (ii) one or more (or all) of the GAG properties selected from the group consisting of: the specific sulfated or unsulfated forms of CS or HS

disaccharides selected from the group consisting of: absolute concentration of 4s CS, absolute concentration of Os CS, relative concentration of Os CS, absolute concentration of Os HS, absolute concentration of Ns HS, absolute concentration of 6s CS, absolute concentration of 2s6s CS, relative concentration of 2s6s CS, relative concentration of 6s CS; CS tot; HS tot; charge CS; or (iii) one or more (or all) of the GAG properties selected from the group consisting of: the specific sulfated or unsulfated forms of CS or HS disaccharides selected from the group consisting of: absolute concentration of 4s CS, relative concentration of Os CS, absolute concentration of Os HS, absolute concentration of Ns HS, absolute concentration of 2s6s CS, relative concentration of 4s CS, relative concentration of 2s6s CS, relative concentration of 6s CS, relative concentration of Os HS, relative concentration of Ns HS; the ratio of 6s CS to Os CS; CS tot; HS tot (specifically the total concentration of HS or CS (see [0311]-[0312]).

Regarding Claim 10, Crawford et al teaches that said body fluid sample is urine (see [0102] and [0106]) and said determination of the chemical composition comprises determining the level in said protein- free fraction of one or more (or all) of the GAG properties selected from the group consisting of: the specific sulfated or unsulfated forms of CS or HS disaccharides selected from the group consisting of: absolute concentration of Os HS, relative concentration of 6s CS, relative concentration of Ns HS, relative concentration of Os HS, relative concentration of 2s6s CS, relative concentration of 4s CS, relative concentration of Os CS, absolute concentration of 4s CS, absolute concentration of 6s CS, absolute concentration of Os CS; the ratio of 6s

CS to Os CS; CS tot; HS tot (specifically the total concentration of HS or CS (see [0311]-[0312]).

Regarding Claim 26, Crawford et al teaches that said level or chemical composition of said GAG or GAG property is determined by HPLC and mass spectrometry (see [0019] and [0125]).

Regarding Claims 29-30, Crawford et al teaches that said subject is a subject at risk of developing cancer, or at risk of the occurrence of cancer, or is a subject having or suspected of having cancer (see [0106]), and wherein said cancer is (i) a stage I cancer, a stage II cancer, a stage III cancer, a stage IV cancer and a cancer of an unspecified stage; or (ii) selected from the group consisting of a genitourinary cancer (e.g. kidney cancer, prostate cancer or bladder cancer), a respiratory tract cancer (e.g. lung cancer), a brain tumor, a blood cancer (e.g. a lymphoma), colorectal cancer, uterine cancer, a gastrointestinal- neuroendocrine tumour, a breast cancer, ovarian cancer and head and neck cancer; or (iii) selected from the group consisting of: bladder cancer, breast invasive ductal carcinoma, cervix squamous cell carcinoma, chronic lymphoid leukaemia, colorectal cancer, endometrial cancer, diffuse glioma, a gastro-intestinal endocrine tumour, head and neck squamous cell carcinoma, diffuse large B-cell lymphoma, non-small cell lung cancer, ovarian cancer, prostate cancer and renal cell cancer (see [0274]-[0280]).

Regarding Claim 31, Crawford et al teaches said method is used for diagnosing cancer, for the prognosis of cancer, for predicting the occurrence of cancer, for estimate the risk of the occurrence of cancer, for monitoring subjects at risk of the occurrence of cancer, for monitoring the progression of cancer in a subject, for determining the clinical

severity of cancer, for predicting the response of a subject to therapy or surgery for cancer, for determining the efficacy of a therapeutic or surgical regime being used to treat cancer, for detecting the recurrence of cancer, for predicting the tissue of origin of a cancer, or for distinguishing small masses suspicious of cancer from other non-malignant diseases (see [0099], [0106] and [0115]).

Claim Rejections - 35 USC § 103

The following is a quotation of 35 U.S.C. 103 which forms the basis for all obviousness rejections set forth in this Office action:

> A patent for a claimed invention may not be obtained, notwithstanding that the claimed invention is not identically disclosed as set forth in section 102, if the differences between the claimed invention and the prior art are such that the claimed invention as a whole would have been obvious before the effective filing date of the claimed invention to a person having ordinary skill in the art to which the claimed invention pertains. Patentability shall not be negated by the manner in which the invention was made.

The factual inquiries for establishing a background for determining obviousness under 35 U.S.C. 103 are summarized as follows:

1. Determining the scope and contents of the prior art.

2. Ascertaining the differences between the prior art and the claims at issue.

3. Resolving the level of ordinary skill in the pertinent art.

4. Considering objective evidence present in the application indicating obviousness or nonobviousness.

Claim(s) 27 and 28 are rejected under 35 U.S.C. 103 as being unpatentable over Crawford et al as applied to claim 26 above, and further in view of Naimy et al (US PGPub 2017/0350900).

Regarding Claim 27, Crawford et al teaches that the level or chemical composition of said GAG or GAG property is determined by HPLC (see [0019]).

However, Crawford et al does not explicitly teach said HPLC is ultra-HPLC.

However, in the analogous art of detecting glycosaminoglycans (GAGs), Naimy et al teaches that the method of separation may include ultra-performance liquid chromatography (UPLC) or high-performance liquid chromatography (HPLC) (see [0126]). It would have been obvious to one of ordinary skill in the art to utilize ultra-HPLC (as taught by Naimy et al) as the type of HPLC utilized since ultra-HPLC is known to separate molecules with effective sensitivity.

Regarding Claim 28, Crawford et al teaches the use of mass spectrometry (see [0019]).

Crawford et al does not explicitly disclose that said mass spectrometry is triple-quadrupole mass spectrometry.

However, in the analogous art of detecting GAGs, Naimy et al teaches that the mass spectrometry analyzer can be selected from any analyzer known in the art, e.g., quadrupole mass analyzers, time of flight mass analyzers, magnetic sector mass analyzers, electrostatic sector mass analyzers, quadrupole ion trap mass analyzers, orbitrap analyzers, or ion cyclotron resonance analyzers (see [0129]). In addition, Naimy et al teaches that subsequent to glycan-specific solid-phase extraction, the disaccharides were analyzed by liquid chromatography in conjunction with tandem mass spectrometry using an API 5000™ triple quadruple mass spectrometer operating in the electrospray negative ion mode (see [0208]). It would have been obvious to one

of ordinary skill in the art to utilize a triple-quadrupole mass spectrometer (as taught by Naimy et al) for the benefit of effectively analyzing the disaccharides.

Claim 27 is rejected under 35 U.S.C. 103 as being unpatentable over Gatto et al as applied to claim 26 above, and further in view of Tanaka et al (US PGPub 2021/0132040).

Regarding Claim 27, Gatto et al teaches using HPLC (see [0045] and [0076]).

However, Gatto does not explicitly disclose that said HPLC is ultra-HPLC.

However, in the analogous art of analytical methods for GAG, Tanaka et al teaches that the high performance liquid chromatography used includes the hydrophilic interaction ultra-high performance liquid chromatography (see [0067] and [0111]). It would have been obvious to one of ordinary skill in the art to utilize ultra-HPLC (as taught by Tanaka et al) for the benefit of effectively retaining disaccharides for analysis.

Claim 28 is rejected under 35 U.S.C. 103 as being unpatentable over Gatto et al as applied to claim 26 above, and further in view of Naimy et al (US PGPub 2017/0350900).

Gatto et al teaches the use of mass spectrometry (see [0045] and [0076]).

Gatto et al does not explicitly disclose that said mass spectrometry is triple-quadrupole mass spectrometry.

However, in the analogous art of detecting GAGs, Naimy et al teaches that the mass spectrometry analyzer can be selected from any analyzer known in the art, e.g., quadrupole mass analyzers, time of flight mass analyzers, magnetic sector mass

analyzers, electrostatic sector mass analyzers, quadrupole ion trap mass analyzers, orbitrap analyzers, or ion cyclotron resonance analyzers (see [0129]). In addition, Naimy et al teaches that subsequent to glycan-specific solid-phase extraction, the disaccharides were analyzed by liquid chromatography in conjunction with tandem mass spectrometry using an API 5000™ triple quadruple mass spectrometer operating in the electrospray negative ion mode (see [0208]). It would have been obvious to one of ordinary skill in the art to utilize a triple-quadrupole mass spectrometer (as taught by Naimy et al) for the benefit of effectively analyzing the disaccharides.

Conclusion

The prior art made of record and not relied upon is considered pertinent to applicant's disclosure:

Gatto et al (US PGPub 2017/0254811) discloses a method of screening for renal cell carcinoma in a subject, said method comprising determining the level and/or chemical composition of one or both of the glycosaminoglycans (GAGs) chondroitin sulfate (CS) and heparan sulfate (HS) in a body fluid sample, wherein said sample has been obtained from said subject.

Ruiz et al (US PGPub 2013/0189718) discloses methods, assays and kits useful for the diagnosis and monitoring of subjects with mucopolysaccharidoses (MPS), The methods, assays and kits are particularly useful for detecting the presence of one or more glycosaminoglycans which correlate to MPS and its severity in a variety of biological samples.

Any inquiry concerning this communication or earlier communications from the examiner should be directed to JENNIFER WECKER whose telephone number is (571)270-1109. The examiner can normally be reached 9:30AM - 6 PM EST M-F.

Examiner interviews are available via telephone, in-person, and video conferencing using a USPTO supplied web-based collaboration tool. To schedule an interview, applicant is encouraged to use the USPTO Automated Interview Request (AIR) at http://www.uspto.gov/interviewpractice.

If attempts to reach the examiner by telephone are unsuccessful, the examiner's supervisor, Lyle Alexander can be reached at 571-272-1254. The fax phone number for the organization where this application or proceeding is assigned is 571-273-8300.

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/JENNIFER WECKER/
Primary Examiner, Art Unit 1797